BUMHEE LEE

Adalbertsteinweg 174, Aachen, Germany, 52066
+49-1763-5849470 | zealatan@gmail.com

PROFESSIONAL SUMMARY

SoC and FPGA-based wireless modem engineer with experience in LTE/NR PHY systems and mmWave prototyping. Skilled in modem implementation and verification using Python, C/C++, and Verilog. Hands-on experience with FPGA platforms, hardware-software integration, and real-time signal processing. Strong understanding of the semiconductor and System-on-Chip industry.

WORKING EXPERIENCE

- **RWTH Aachen University** **Aachen, Germany**
  Research Assistant **– Chair for Distributed Signal Processing** *Aug 2023 – Aug 2026*
  - Developed multi-rate, Gbps high-speed OFDM transceiver systems on RFSoC.
  - Designed AI-native neural synchronizer on SDR.
  - Built a programmable 28 GHz MU-MIMO testbed with real-time beam tracking.

- **Samsung Electronics (S.LSI)** **Hwaseong, South Korea**
  Algorithm Researcher **– Next Generation Task Force** *Nov 2017 – May 2018*
  - Researched memory-efficient neural network optimization techniques for NPUs, including NAS, quantization, and pruning.
  - Conducted model compression for reducing inference memory footprint.
  - Generated fixed point test vectors for NPU verification.

  Software Engineer **– Protocol team** *Sep 2015 – Oct 2017*
  - Participated in HRPD (High-Rate Packet Data) protocol development.
  - Designed Layer 1 downlink modules for mobility and cell management based on 3GPP2 C.S0074.
  - Resolved Layer 1 issues related to closed-loop power control, HARQ, and symbol detection.

  FPGA Test Engineer **– Multimedia Core team** *Mar 2013 – Aug 2015*
  - Performed functional verification of multi-format video codec IPs on FPGA (HAPs).
  - Executed over 100,000 test cases for VP9, MPEG, HEVC, and H.264 codecs.
  - Contributed to test automation that reduced overall test time by up to 15%.

PUBLICATIONS

- First author of "Programmable 28 GHz mmWave MU-MIMO Testbed," *Proc. IEEE WCNC 2025*, Milan, Italy; demonstrated a real-time 28 GHz MU-MIMO system.

EDUCATION

- **Friedrich-Alexander-Universität Erlangen-Nürnberg** **Erlangen, Germany**
  M.Sc. in Communication and Multimedia Engineering *Oct 2019 – Dec 2022*
  - Developed power-efficient reflected intelligent surface (RIS)-assisted WPT algorithms achieving ~11% transmission power reduction in simulation for Master thesis

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- **Hanyang University** **Seoul, South Korea**
  B.Sc. in Electronics and Computer Engineering *Mar 2006 – Feb 2013*
  - Completed mandatory Military service (2008 – 2010)

TECHNICAL SKILLS

Programming
C/C++, Python, Verilog/SystemVerilog

FPGA/Hardware Platforms
Xilinx Vivado/Vitis, RFSoC (ZCU216), USRP X410

System Integration
ARM architecture, AXI BUS, DMA, streaming dataflow design

Tools & Measurement Equipment
Trace32, Git, Spectrum Analyzer, Oscilloscope

Communication Systems
LTE/NR, EVDO, Real-time PHY implementation

LANGUAGE

Korean: Native
English: Professional working proficiency (C1)
German: Basic (A2)